

10/7 H.W

- 1) → Find out the History of Towers of Hanoi.
- 2) → Today, from the morning 7:30 AM to 4:30 PM
List out various Data structures, you
observed in your surroundings

1) History of Towers of Hanoi: -

The Tower of Hanoi puzzle was invented by a French Mathematician named Edouard Lucas in 1883. Here's a

Myth Attached to it about Brahmin priests in a Temple moving 64 Golden discs between three pillars, & the legend says when they finish, the world will end. In computer science, we use it mainly to understand & practice Recursion, since the problem breaks down into smaller repeating sub-

problems.

2) Data structures observation (7:30 AM - 9:30 AM)

1) Stack :- The pile of plates in kitchen at breakfast, plates are added & removed from top only (Last In - first out).

2) Queue :- Students standing in line at to get ~~down~~^{into} college bus. (First In - First out).

3) Array :- The rows of desks in classes - an index-like sequence - fixed).

4) Linked-list :- Auto 57 Buses lined up one behind another on road, each one connected to the next.

5) Tree :- The folder structure in Lab room, main folder → sub-folder → then files inside them.

6) Graph :- My whatsapp ~~gr~~ Bus group shared Bus's live location on the Morning as daily, they do. So, as it moves from stop to stop, each stop is like a node & road between 2 stops is like an edge connecting them. As Bus moves, it's like moving from 1 Node to Another, just like in a Graph.